

ML2025 Fall Homework Assignment 3

Handwritten

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Tools You Need to Know/Learn

- Data centering, Covariance matrix
- Multivariate Gaussian, KL-Divergence
- Graph construction, Laplacian
- EM Algorithm: E-step, M-step updates

You are expected to know (or learn through this homework) the definitions and usages of the above concepts, and are encouraged to discuss with TAs during TA hour if you encounter difficulties.

Homework Policy

- The homework is graded out of 105 points.
- The official submission deadline will follow the schedule announced on NTU COOL.
- Homework may be handwritten or typed (e.g. using $\text{\LaTeX}$), but must be submitted in **PDF format**.
- If you discuss the homework with classmates, you should state their student IDs in your submission.
- Plagiarism is strictly prohibited. Serious violations will be dealt with according to NTU regulations.

Problem 1 (Laplacian Eigenmaps) (25 pts)

Consider an undirected connected graph G , which is shown below. We want to utilize Laplacian Eigenmaps method to reduce these 10 points to 3-dimensional space. Here, undirected graph means that edges in the graph do not have a direction, and connected graph means that there is a path from any node to any other node in the graph.

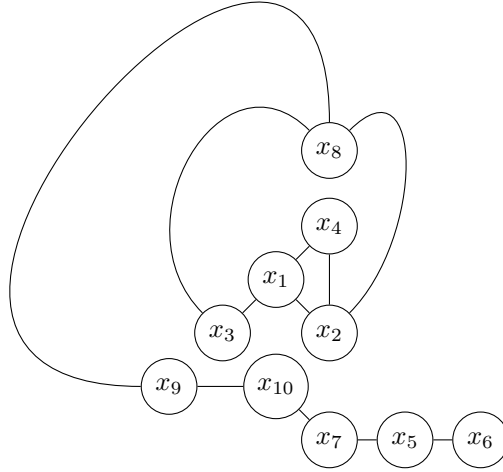


Figure 1: undirected connected graph G

1. (2 pts) Write down the adjacency matrix $\mathbf{W}$
2. (2 pts) Write down the diagonal matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_{10})$, where $d_i = \sum_{j=1}^{10} \frac{\mathbf{W}_{ij} + \mathbf{W}_{ji}}{2}$ and the Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{W}$.
3. (5 pts) Based on Neighbor Embedding Slide p.7-p.10, use programming tools (e.g., MATLAB or Python) to implement and solve the following optimization problem.

$$\begin{aligned}
 &\text{minimize} && \text{tr}(\mathbf{\Psi}^\top \mathbf{L} \mathbf{\Psi}) \\
 &\text{subject to} && \mathbf{\Psi}^\top \mathbf{D} \mathbf{\Psi} = \mathbf{I}_3 \\
 &\text{variables} && \mathbf{\Psi} \in \mathbb{R}^{10 \times 3}
 \end{aligned}$$

Include your code in [this file](#) using either (i) Overleaf's `minted` environment (compile with `-shell-escape`) or (ii) a styled screenshot generated with Carbon (<https://carbon.now.sh/>). Also, please plot the reduced points $\mathbf{z}_1, \dots, \mathbf{z}_{10}$ in a 3-D scatter plot.

4. (2 pts) You may find that the minimal eigenvalue of $\mathbf{L}$ is 0, and the corresponding eigenvector is

$$\begin{bmatrix} c \\ c \\ \vdots \\ c \end{bmatrix} \quad (1)$$

where c is a constant. Since all the points fall into a plane, the span of these points is $\mathbb{R}^2$. In order to construct $\mathbf{z}_1, \dots, \mathbf{z}_{10}$ such that $\text{span}\{\mathbf{z}_1, \dots, \mathbf{z}_{10}\} = \mathbb{R}^3$, we need choose the second, third, fourth smallest eigenvalue and the corresponding eigenvectors. Please plot the reduced points by the updated $\mathbf{z}_1, \dots, \mathbf{z}_{10}$ in 3-D scatter plot and verify that whether $\text{tr}(\mathbf{\Psi}^\top \mathbf{L} \mathbf{\Psi}) = 1.098$ and $\mathbf{\Psi}^\top \mathbf{D} \mathbf{\Psi} = \mathbf{I}_3$.

5. (3 pts) Show that for no matter the graph is, there is an eigenvector of $\mathbf{L}$

$$\begin{bmatrix} c \\ c \\ \vdots \\ c \end{bmatrix} \quad (2)$$

where c is a constant, and the corresponding eigenvalue is 0.

6. (3 pts) By Neighbor Embedding Slide p.9, please show that

$$\mathbf{f}^\top \mathbf{L} \mathbf{f} = \frac{1}{2} \sum_{1 \leq i, j \leq N} w_{ij} (f_i - f_j)^2, \quad \forall \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix} \in \mathbb{R}^N.$$

7. (3 pts) Show that if $\mathbf{f}$ is an eigenvector of $\mathbf{L}$ which corresponds to eigenvalue 0, then $\mathbf{f}^\top \mathbf{L} \mathbf{f} = 0$.
8. (5 pts) Show that if the graph is connected, the second smallest eigenvalue of $\mathbf{L}$ will be nonzero.

Solution: (a) By reading off the edges in the graph, the adjacency matrix $\mathbf{W} \in \mathbb{R}^{10 \times 10}$ (ordering vertices as $x_1, \dots, x_{10}$) is

$$\mathbf{W} = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}.$$

(b) Each degree is $d_i = \sum_{j=1}^{10} W_{ij}$, so

$$(d_1, \dots, d_{10}) = (3, 3, 2, 2, 2, 1, 2, 3, 2, 2).$$

Hence

$$\mathbf{D} = \text{diag}(3, 3, 2, 2, 2, 1, 2, 3, 2, 2),$$

and the (unnormalized) graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{W}$ is

$$\mathbf{L} = \begin{bmatrix} 3 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 3 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & -1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 2 & 0 & 0 & -1 \\ 0 & -1 & -1 & 0 & 0 & 0 & 0 & 3 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 2 \end{bmatrix}.$$

(c) We want to solve

$$\min_{\Psi \in \mathbb{R}^{10 \times 3}} \text{tr}(\Psi^\top \mathbf{L} \Psi) \quad \text{s.t.} \quad \Psi^\top \mathbf{D} \Psi = \mathbf{I}_3.$$

This is a generalized eigenvalue problem:

$$\mathbf{L} \psi_k = \lambda_k \mathbf{D} \psi_k,$$

and the optimizer Ψ is formed by stacking as columns the three generalized eigenvectors corresponding to the three smallest eigenvalues.

Why does the Laplacian Eigenmaps objective lead to eigenvectors? Write $\Psi = [\psi_1, \dots, \psi_k]$. Then

$$\text{tr}(\Psi^\top \mathbf{L} \Psi) = \sum_{j=1}^k \psi_j^\top \mathbf{L} \psi_j, \quad \Psi^\top \mathbf{D} \Psi = \mathbf{I}_k \iff \psi_i^\top \mathbf{D} \psi_j = \delta_{ij}.$$

Thus we are choosing k $\mathbf{D}$ -orthonormal vectors to minimize $\sum_j \psi_j^\top \mathbf{L} \psi_j$. Introducing a matrix Lagrange multiplier Λ and differentiating

$$\mathcal{L}(\Psi, \Lambda) = \text{tr}(\Psi^\top \mathbf{L} \Psi) - \text{tr}(\Lambda(\Psi^\top \mathbf{D} \Psi - \mathbf{I}_k))$$

with respect to Ψ yields the stationarity condition

$$\mathbf{L} \Psi = \mathbf{D} \Psi \Lambda.$$

Hence each column ψ_j satisfies $\mathbf{L}\psi_j = \lambda_j \mathbf{D}\psi_j$, i.e. it is a generalized eigenvector of $(\mathbf{L}, \mathbf{D})$. Moreover, $\text{tr}(\mathbf{\Psi}^\top \mathbf{L} \mathbf{\Psi}) = \sum_{j=1}^k \lambda_j$, so the minimum is achieved by taking the k smallest eigenvalues (and their associated eigenvectors).

In (e.g.) Python, one can do:

```
import numpy as np
from scipy.linalg import eigh

W = np.array([...]) # 10x10 matrix above
D = np.diag(W.sum(axis=1))
L = D - W

# Solve L v = lambda D v
evals, evects = eigh(L, D)

# Take 3 smallest eigenvalues/eigenvectors
idx = np.argsort(evals)
Psi = evects[:, idx[:3]] # shape (10, 3)

# Each row z_i = Psi[i, :] in R^3 is the embedded point
```

Then we can plot the reduced points $\mathbf{z}_i \in \mathbb{R}^3$ using any 3D scatter function, e.g. `matplotlib's Axes3D.scatter`.

(d) The smallest generalized eigenvalue is $\lambda_1 = 0$, with eigenvector proportional to the constant vector

$$\mathbf{v}_1 = \begin{bmatrix} c \\ c \\ c \\ \vdots \\ c \end{bmatrix}.$$

If we use this eigenvector as one column of $\mathbf{\Psi}$, all embedded points will lie in a 2D subspace (a plane), so their span is only $\mathbb{R}^2$. To obtain an embedding whose span is $\mathbb{R}^3$, we instead choose the eigenvectors corresponding to the second, third, and fourth smallest eigenvalues:

$$\mathbf{\Psi} = [\psi_2 \quad \psi_3 \quad \psi_4].$$

Numerically, using the same code as above, we simply change the indices:

```
# Skip the eigenvector for eigenvalue 0
Psi = evects[:, idx[1:4]] # columns for lambda_2, lambda_3, lambda_4

# Verify constraints numerically
val = np.trace(Psi.T @ L @ Psi) # should be close to 1.098 (given in HW)
orth = Psi.T @ D @ Psi # should be close to I_3
```

The rows of this new $\mathbf{\Psi}$ again give the embedded points $\mathbf{z}_1, \dots, \mathbf{z}_{10} \in \mathbb{R}^3$, and one can check that

$$\text{tr}(\mathbf{\Psi}^\top \mathbf{L} \mathbf{\Psi}) \approx 1.098, \quad \mathbf{\Psi}^\top \mathbf{D} \mathbf{\Psi} \approx \mathbf{I}_3$$

up to numerical precision.

(e) Let

$$\mathbf{v} = \begin{bmatrix} c \\ c \\ \vdots \\ c \end{bmatrix} \in \mathbb{R}^N.$$

Since $\mathbf{L} = \mathbf{D} - \mathbf{W}$ and $D_{ii} = \sum_j W_{ij}$, we have, for each coordinate i ,

$$(\mathbf{L}\mathbf{v})_i = \sum_j L_{ij}v_j = \sum_j (D_{ij} - W_{ij})c = c\left(D_{ii} - \sum_j W_{ij}\right) = c(\deg(i) - \deg(i)) = 0.$$

Thus $\mathbf{L}\mathbf{v} = \mathbf{0} = 0 \cdot \mathbf{v}$, proving that $\mathbf{v}$ is an eigenvector of $\mathbf{L}$ with eigenvalue 0, for any graph.

(f) Let $\mathbf{f} = (f_1, \dots, f_N)^\top \in \mathbb{R}^N$. Then

$$\mathbf{f}^\top \mathbf{L}\mathbf{f} = \mathbf{f}^\top \mathbf{D}\mathbf{f} - \mathbf{f}^\top \mathbf{W}\mathbf{f} = \sum_i d_i f_i^2 - \sum_{i,j} w_{ij} f_i f_j,$$

where $d_i = \sum_j w_{ij}$ and $w_{ij} = w_{ji}$. Write

$$\sum_i d_i f_i^2 = \sum_i \left(\sum_j w_{ij} \right) f_i^2 = \sum_{i,j} w_{ij} f_i^2.$$

Hence

$$\mathbf{f}^\top \mathbf{L}\mathbf{f} = \sum_{i,j} w_{ij} f_i^2 - \sum_{i,j} w_{ij} f_i f_j.$$

Similarly, by symmetry $w_{ij} = w_{ji}$,

$$\sum_{i,j} w_{ij} f_j^2 = \sum_{i,j} w_{ij} f_i^2,$$

so

$$\sum_{i,j} w_{ij} (f_i - f_j)^2 = \sum_{i,j} w_{ij} (f_i^2 + f_j^2 - 2f_i f_j) = 2 \sum_{i,j} w_{ij} f_i^2 - 2 \sum_{i,j} w_{ij} f_i f_j = 2 \mathbf{f}^\top \mathbf{L}\mathbf{f}.$$

Therefore,

$$\boxed{\mathbf{f}^\top \mathbf{L}\mathbf{f} = \frac{1}{2} \sum_{1 \leq i,j \leq N} w_{ij} (f_i - f_j)^2}.$$

(g) If $\mathbf{f}$ is an eigenvector of $\mathbf{L}$ with eigenvalue 0, then

$$\mathbf{L}\mathbf{f} = 0 \cdot \mathbf{f} = \mathbf{0},$$

and hence

$$\mathbf{f}^\top \mathbf{L}\mathbf{f} = \mathbf{f}^\top \mathbf{0} = 0.$$

Alternatively, by the formula in (f),

$$0 = \mathbf{f}^\top \mathbf{L}\mathbf{f} = \frac{1}{2} \sum_{i,j} w_{ij} (f_i - f_j)^2,$$

and since each term $w_{ij} (f_i - f_j)^2 \geq 0$, we must have $f_i = f_j$ whenever $w_{ij} > 0$.

(h) Assume the graph is connected. From part (f),

$$\mathbf{f}^\top \mathbf{L}\mathbf{f} = \frac{1}{2} \sum_{i,j} w_{ij} (f_i - f_j)^2 \geq 0,$$

so all eigenvalues of $\mathbf{L}$ are nonnegative (i.e. $\mathbf{L}$ is positive semidefinite). From part (e), the constant vector $\mathbf{1} = (1, \dots, 1)^\top$ is an eigenvector with eigenvalue 0, so 0 is the smallest eigenvalue.

Suppose, for contradiction, that the second smallest eigenvalue λ_2 is also 0. Then there exists another eigenvector $\mathbf{g}$ linearly independent from $\mathbf{1}$ such that

$$L\mathbf{g} = 0 \cdot \mathbf{g} = \mathbf{0}.$$

By part (f), this implies

$$\sum_{i,j} w_{ij} (g_i - g_j)^2 = 0 \quad \Rightarrow \quad g_i = g_j \text{ whenever } w_{ij} > 0.$$

Since the graph is connected, any two vertices are linked by a path of edges with $w_{ij} > 0$, so all g_i must be equal; that is, $\mathbf{g}$ must be a constant vector. This contradicts the assumption that $\mathbf{g}$ is linearly independent of $\mathbf{1}$.

Therefore, the eigenspace corresponding to eigenvalue 0 is one-dimensional (spanned by the constant vector), and all other eigenvalues must be strictly positive. In particular, the second smallest eigenvalue λ_2 satisfies

$$\lambda_2 > 0$$

whenever the graph is connected.

Problem 2 (Principal Component Analysis) (20 pts)

Let $(\mathbf{x}_i)_{i=1}^N$ be N centered data points in $\mathbb{R}^d$ (i.e., $\sum_{i=1}^N \mathbf{x}_i = \mathbf{0}$). Denote the empirical covariance

$$\mathbf{\Sigma} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^\top \in \mathbb{R}^{d \times d}.$$

Throughout, assume $\mathbf{\Sigma}$ is symmetric positive semidefinite.

- (a) (5 pts) Show that the first principal axis $\mathbf{u}_1$ satisfies

$$\mathbf{u}_1 = \arg \max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u},$$

and that any maximizer must satisfy the eigenvalue equation $\mathbf{\Sigma} \mathbf{u}_1 = \lambda_1 \mathbf{u}_1$ with $\lambda_1 = \max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}$.

- (b) (5 pts) Let $\hat{\mathbf{x}}_i = (\mathbf{u}^\top \mathbf{x}_i) \mathbf{u}$ be the orthogonal projection of $\mathbf{x}_i$ onto the line spanned by a unit vector $\mathbf{u}$.

Prove the identity

$$\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = \text{tr}(\mathbf{\Sigma}) - \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u},$$

and conclude that minimizing the reconstruction error over unit $\mathbf{u}$ is equivalent to part (a).

- (c) (5+5 pts) Now take $d = 2$ (i.e., $\mathbf{x}_i = (x_{i,1}, x_{i,2})$) and suppose you are given the aggregated statistics

$$\sum_{i=1}^N x_{i,1} = \sum_{i=1}^N x_{i,2} = 0, \quad \sum_{i=1}^N x_{i,1}^2 = 363, \quad \sum_{i=1}^N x_{i,1} x_{i,2} = -60, \quad \sum_{i=1}^N x_{i,2}^2 = 482.$$

- (i) Compute the eigenvalues and a unit eigenvector $\mathbf{u}_1$ corresponding to the largest eigenvalue.
- (ii) Using part (b), express the total reconstruction error $\sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2$ along $\mathbf{u}_1$ in closed form.

Solution:

(a) For any unit vector $\mathbf{u}$, the variance of the projection $\mathbf{u}^\top \mathbf{x}_i$ is

$$\frac{1}{N} \sum_{i=1}^N (\mathbf{u}^\top \mathbf{x}_i)^2 = \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}.$$

Thus the first principal axis is defined as

$$\mathbf{u}_1 = \arg \max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}.$$

Introducing a Lagrange multiplier λ for $\|\mathbf{u}\|_2 = 1$, we consider

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1).$$

Setting $\nabla_{\mathbf{u}} \mathcal{L} = 0$ yields

$$\mathbf{\Sigma} \mathbf{u} = \lambda \mathbf{u}.$$

Hence any maximizer must be an eigenvector of $\mathbf{\Sigma}$, and the maximized Rayleigh quotient is the largest eigenvalue:

$$\lambda_1 = \max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}.$$

(b) The projection of $\mathbf{x}_i$ onto the unit direction $\mathbf{u}$ is

$$\hat{\mathbf{x}}_i = (\mathbf{u}^\top \mathbf{x}_i) \mathbf{u}.$$

Then

$$\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = \|\mathbf{x}_i\|_2^2 - (\mathbf{u}^\top \mathbf{x}_i)^2.$$

Averaging:

$$\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i\|_2^2 - \frac{1}{N} \sum_{i=1}^N (\mathbf{u}^\top \mathbf{x}_i)^2.$$

Since

$$\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^\top = \mathbf{\Sigma}, \quad \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i\|_2^2 = \text{tr}(\mathbf{\Sigma}),$$

we obtain

$$\boxed{\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = \text{tr}(\mathbf{\Sigma}) - \mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}.}$$

Since $\text{tr}(\mathbf{\Sigma})$ is constant, minimizing reconstruction error is equivalent to maximizing $\mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}$, which is exactly part (a).

(c) Given:

$$\sum x_{i,1}^2 = 363, \quad \sum x_{i,1} x_{i,2} = -60, \quad \sum x_{i,2}^2 = 482,$$

the (unnormalized) covariance matrix is

$$\mathbf{S} = \begin{bmatrix} 363 & -60 \\ -60 & 482 \end{bmatrix}, \quad \mathbf{\Sigma} = \frac{1}{N} \mathbf{S}.$$

(i) **Eigenvalues and eigenvector.** Solve

$$\det(\mathbf{S} - \mu \mathbf{I}) = 0.$$

This yields eigenvalues

$$\mu_1 = 507, \quad \mu_2 = 338,$$

hence

$$\lambda_1 = \frac{507}{N}, \quad \lambda_2 = \frac{338}{N}.$$

For $\mu_1 = 507$,

$$(\mathbf{S} - 507\mathbf{I}) = \begin{bmatrix} -144 & -60 \\ -60 & -25 \end{bmatrix},$$

which gives the relation

$$\mathbf{u}_1 = -\frac{5}{12}\mathbf{u}_2.$$

A normalized eigenvector is therefore

$$\mathbf{u}_1 = \begin{bmatrix} -5/13 \\ 12/13 \end{bmatrix}.$$

(ii) Total reconstruction error. From part (b),

$$\sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = N(\text{tr}(\mathbf{\Sigma}) - \lambda_1) = 845 - 507.$$

Thus

$$\sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 = 338.$$

Problem 3 (Gradient of the t-SNE Objective) (10pts)

By working on this problem, we hope that students will gain a deeper understanding of the mathematical formulation of the t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm, particularly the derivation of its gradient from the Kullback–Leibler (KL) divergence objective.

The goal of t-SNE is to find low-dimensional representations $\{\mathbf{y}_i\}_{i=1}^N$ that preserve local similarities of high-dimensional data $\{\mathbf{x}_i\}_{i=1}^N$.

We define the pairwise *high-dimensional similarities* as

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad \text{where} \quad p_{j|i} = \frac{\exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_k\|^2}{2\sigma_i^2}\right)}.$$

The *low-dimensional similarities* are modeled using a Student- t kernel with one degree of freedom:

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq \ell} (1 + \|\mathbf{y}_k - \mathbf{y}_\ell\|^2)^{-1}}, \quad \text{and} \quad q_{ii} = 0.$$

The symmetric t-SNE objective minimizes the KL divergence between these two distributions:

$$C = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

1. (2 pts) Prove that the cost can be simplified to

$$C = \text{constant}(\text{to } \mathbf{y}_i) - \sum_{i \neq j} p_{ij} \log q_{ij}.$$

2. (8 pts) Show that the final expression of the gradient is

$$\frac{\partial C}{\partial \mathbf{y}_i} = 4 \sum_{j \neq i} (p_{ij} - q_{ij})(\mathbf{y}_i - \mathbf{y}_j)(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}.$$

Hint: Substitute the definition of q_{ij} into the cost function, and you may use the following useful hint:

$$\log\left(\frac{A}{B}\right) = \log A - \log B, \quad \frac{\partial \log(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)}{\partial \mathbf{y}_i} = \frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Carefully account for the contributions where $\mathbf{y}_i$ appears in both indices of the summation.

Solution.

(1) Simplifying the cost function

The t-SNE objective is

$$C = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

Using the identity $\log \frac{A}{B} = \log A - \log B$, we obtain

$$\begin{aligned} C &= \sum_{i \neq j} p_{ij} (\log p_{ij} - \log q_{ij}) \\ &= \underbrace{\sum_{i \neq j} p_{ij} \log p_{ij}}_{\text{depends only on } p_{ij}, \text{ fixed}} - \sum_{i \neq j} p_{ij} \log q_{ij}. \end{aligned}$$

The high-dimensional similarities p_{ij} are fixed once the $\mathbf{x}_i$ and the bandwidths σ_i are chosen, so

$$\sum_{i \neq j} p_{ij} \log p_{ij}$$

does not depend on the low-dimensional variables $\{\mathbf{y}_i\}$. Therefore, with respect to the optimization over $\{\mathbf{y}_i\}$, this term is a constant. Hence we can write

$$C = \text{constant (in } \mathbf{y}_i) - \sum_{i \neq j} p_{ij} \log q_{ij}.$$

(2) Deriving the gradient $\partial C / \partial \mathbf{y}_i$

We now derive the gradient of

$$C' := - \sum_{i \neq j} p_{ij} \log q_{ij},$$

noting that the additive constant from part (1) has zero gradient.

Step 1: Rewrite q_{ij} with an explicit normalization

Define

$$w_{ij} := (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}, \quad \text{for } i \neq j, \quad \text{and} \quad Z := \sum_{k \neq \ell} w_{k\ell}.$$

Then

$$q_{ij} = \frac{w_{ij}}{Z}, \quad q_{ii} = 0.$$

Using this in C' ,

$$\begin{aligned} C' &= - \sum_{i \neq j} p_{ij} \log \left(\frac{w_{ij}}{Z} \right) \\ &= - \sum_{i \neq j} p_{ij} (\log w_{ij} - \log Z) \\ &= - \sum_{i \neq j} p_{ij} \log w_{ij} + \left(\sum_{i \neq j} p_{ij} \right) \log Z. \end{aligned}$$

Since $\{p_{ij}\}_{i \neq j}$ is a probability distribution over pairs (i, j) ,

$$\sum_{i \neq j} p_{ij} = 1,$$

so we can write

$$C' = - \sum_{i \neq j} p_{ij} \log w_{ij} + \log Z.$$

Thus, up to the same additive constant as before, we can take

$$C(\mathbf{Y}) = - \sum_{i \neq j} p_{ij} \log w_{ij} + \log Z.$$

Step 2: Differentiate the first term

We first compute the gradient of

$$C_1 := - \sum_{i \neq j} p_{ij} \log w_{ij}.$$

Note that

$$w_{ij} = (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}, \quad \log w_{ij} = -\log(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2).$$

By the hint,

$$\frac{\partial}{\partial \mathbf{y}_i} \log(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2) = \frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Therefore,

$$\frac{\partial}{\partial \mathbf{y}_i} \log w_{ij} = -\frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Now, $\mathbf{y}_i$ appears in all terms with index pairs (i, j) and (j, i) in the sum $\sum_{i \neq j}$. Hence

$$\frac{\partial C_1}{\partial \mathbf{y}_i} = -\sum_{j \neq i} p_{ij} \frac{\partial}{\partial \mathbf{y}_i} \log w_{ij} - \sum_{k \neq i} p_{ki} \frac{\partial}{\partial \mathbf{y}_i} \log w_{ki}.$$

Using symmetry $w_{ij} = w_{ji}$ and the derivative above,

$$\frac{\partial}{\partial \mathbf{y}_i} \log w_{ij} = \frac{\partial}{\partial \mathbf{y}_i} \log w_{ji} = -\frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2},$$

we obtain

$$\begin{aligned} \frac{\partial C_1}{\partial \mathbf{y}_i} &= -\sum_{j \neq i} p_{ij} \left(-\frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2} \right) - \sum_{j \neq i} p_{ji} \left(-\frac{2(\mathbf{y}_i - \mathbf{y}_j)}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2} \right) \\ &= 2 \sum_{j \neq i} (p_{ij} + p_{ji}) \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}. \end{aligned}$$

In t-SNE, p_{ij} is symmetric, $p_{ij} = p_{ji}$, so

$$p_{ij} + p_{ji} = 2p_{ij}.$$

Therefore

$$\boxed{\frac{\partial C_1}{\partial \mathbf{y}_i} = 4 \sum_{j \neq i} p_{ij} \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Step 3: Differentiate the normalization term $\log Z$

Recall

$$C_2 := \log Z, \quad Z = \sum_{k \neq \ell} w_{k\ell}.$$

By the chain rule,

$$\frac{\partial C_2}{\partial \mathbf{y}_i} = \frac{1}{Z} \frac{\partial Z}{\partial \mathbf{y}_i}.$$

We have

$$\frac{\partial Z}{\partial \mathbf{y}_i} = \sum_{k \neq \ell} \frac{\partial w_{k\ell}}{\partial \mathbf{y}_i}.$$

Only the pairs (i, j) and (j, i) depend on $\mathbf{y}_i$, so

$$\frac{\partial Z}{\partial \mathbf{y}_i} = \sum_{j \neq i} \frac{\partial w_{ij}}{\partial \mathbf{y}_i} + \sum_{j \neq i} \frac{\partial w_{ji}}{\partial \mathbf{y}_i}.$$

We compute

$$w_{ij} = (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1},$$

so by the chain rule

$$\frac{\partial w_{ij}}{\partial \mathbf{y}_i} = -(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-2} \cdot 2(\mathbf{y}_i - \mathbf{y}_j) = -\frac{2(\mathbf{y}_i - \mathbf{y}_j)}{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^2}.$$

Using symmetry $w_{ij} = w_{ji}$, the same formula holds for $\partial w_{ji}/\partial \mathbf{y}_i$, and thus

$$\frac{\partial Z}{\partial \mathbf{y}_i} = -4 \sum_{j \neq i} \frac{(\mathbf{y}_i - \mathbf{y}_j)}{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^2}.$$

Hence

$$\frac{\partial C_2}{\partial \mathbf{y}_i} = \frac{1}{Z} \frac{\partial Z}{\partial \mathbf{y}_i} = -\frac{4}{Z} \sum_{j \neq i} \frac{\mathbf{y}_i - \mathbf{y}_j}{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^2}.$$

Now notice that

$$w_{ij} = \frac{1}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}, \quad q_{ij} = \frac{w_{ij}}{Z}.$$

Thus

$$\frac{w_{ij}^2}{Z} = \frac{1}{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^2} \frac{1}{Z} = \frac{q_{ij}}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Since

$$\frac{\mathbf{y}_i - \mathbf{y}_j}{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^2} = w_{ij}^2 (\mathbf{y}_i - \mathbf{y}_j),$$

we can rewrite

$$\frac{\partial C_2}{\partial \mathbf{y}_i} = -4 \sum_{j \neq i} \frac{w_{ij}^2}{Z} (\mathbf{y}_i - \mathbf{y}_j) = -4 \sum_{j \neq i} \frac{q_{ij}}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2} (\mathbf{y}_i - \mathbf{y}_j).$$

Hence

$$\boxed{\frac{\partial C_2}{\partial \mathbf{y}_i} = -4 \sum_{j \neq i} q_{ij} \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}.$$

Step 4: Combine both parts

The full cost (up to constants) was

$$C = C_1 + C_2 = - \sum_{i \neq j} p_{ij} \log w_{ij} + \log Z.$$

Therefore the gradient is

$$\frac{\partial C}{\partial \mathbf{y}_i} = \frac{\partial C_1}{\partial \mathbf{y}_i} + \frac{\partial C_2}{\partial \mathbf{y}_i}.$$

Using the expressions derived above,

$$\begin{aligned} \frac{\partial C}{\partial \mathbf{y}_i} &= 4 \sum_{j \neq i} p_{ij} \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2} - 4 \sum_{j \neq i} q_{ij} \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2} \\ &= 4 \sum_{j \neq i} (p_{ij} - q_{ij}) \frac{\mathbf{y}_i - \mathbf{y}_j}{1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2}. \end{aligned}$$

Thus we obtain the desired final form:

$$\boxed{\frac{\partial C}{\partial \mathbf{y}_i} = 4 \sum_{j \neq i} (p_{ij} - q_{ij}) (\mathbf{y}_i - \mathbf{y}_j) (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}.$$

Problem 4 (EM for Mixture of Multivariate t -Distributions) (30 pts)

Definitions (first mention). When the Gamma function and Gamma distribution are first used here, adopt the following standard definitions:

- **Gamma function:** for $a > 0$,

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt.$$

- **Gamma distribution (shape–rate parametrization):** a random variable U has $\text{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$ and rate $\beta > 0$ if its density is

$$f_U(u) = \frac{\beta^\alpha}{\Gamma(\alpha)} u^{\alpha-1} e^{-\beta u}, \quad u > 0.$$

(Note you may solve below problem without knowing the above definition, but it's good to know that.)

- **Digamma and trigamma:** the digamma function $\psi(\cdot)$ is the logarithmic derivative of the Gamma function,

$$\psi(a) = \frac{d}{da} \log \Gamma(a) = \frac{\Gamma'(a)}{\Gamma(a)},$$

and the trigamma $\psi'(\cdot)$ is its derivative, $\psi'(a) = \frac{d}{da} \psi(a)$.

Consider the generative model parameterized by

$$\theta = ((\pi_k, \boldsymbol{\mu}_k, \Sigma_k, \nu_k))_{k=1}^K,$$

where each component is a p -variate Student- t distribution with center $\boldsymbol{\mu}_k \in \mathbb{R}^p$, positive-definite scale matrix $\Sigma_k \in \mathbb{R}^{p \times p}$, degrees of freedom $\nu_k > 0$, mixing weight $\pi_k > 0$, and $\sum_{k=1}^K \pi_k = 1$.

For an observation $\mathbf{y} \in \mathbb{R}^p$ the mixture density is

$$p_\theta(\mathbf{y}) = \sum_{k=1}^K \pi_k t_p(\mathbf{y}; \boldsymbol{\mu}_k, \Sigma_k, \nu_k),$$

where the multivariate Student- t pdf (as in Liu & Rubin (1995)) is

$$t_p(\mathbf{y}; \boldsymbol{\mu}, \Sigma, \nu) = \frac{\Gamma(\frac{\nu+p}{2})}{\Gamma(\frac{\nu}{2})(\nu\pi)^{p/2}|\Sigma|^{1/2}} \left(1 + \frac{1}{\nu}(\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu})\right)^{-\frac{\nu+p}{2}}.$$

Suppose we observe N i.i.d. samples $\mathbf{y}_1, \dots, \mathbf{y}_N$. The maximum likelihood estimator is

$$\theta^{\text{opt}} = \arg \max_{\theta} \prod_{i=1}^N p_\theta(y_i).$$

The following problems lead you to derive the E-step and M-step equations of the EM algorithm for mixture of multivariate t -distributions.

EM Algorithm — E-step — posterior expectations and responsibilities.

As mentioned in class, we can rewrite the MLE as

$$\theta^{\text{opt}} = \arg \max_{\theta} (Q(\theta \mid \theta^{(t)}) + H(\theta \mid \theta^{(t)}))$$

by introducing the latent labels $z_i \in \{1, \dots, K\}$.

1. (5 pts) Derive the $Q(\theta \mid \theta^{(t)})$ and $H(\theta \mid \theta^{(t)})$. (You can denote $\frac{\pi_k^{(t)} t_p(\mathbf{y}_i; \boldsymbol{\mu}_k^{(t)}, \Sigma_k^{(t)}, \nu_k^{(t)})}{\sum_{\ell=1}^K \pi_{\ell}^{(t)} t_p(\mathbf{y}_i; \boldsymbol{\mu}_{\ell}^{(t)}, \Sigma_{\ell}^{(t)}, \nu_{\ell}^{(t)})}$ as $\delta_{i,k}^{(t)}$)
2. (4 pts)(you may direct use the result although you do not prove it) if we introduce the latent scale variables $u_{i,k} > 0$ via the standard scale-mixture representation of the multivariate t , we are given

$$\mathbf{y}_i \mid (z_i = k, u_{i,k}) \sim \mathcal{N}\left(\boldsymbol{\mu}_k, \frac{\Sigma_k}{u_{i,k}}\right), \quad u_{i,k} \mid (z_i = k) \sim \text{Gamma}\left(\frac{\nu_k}{2}, \frac{\nu_k}{2}\right),$$

where the Gamma is parameterized by (shape, rate), and define the Mahalanobis distance

$$d_{i,k}(\boldsymbol{\mu}_k, \Sigma_k) = (\mathbf{y}_i - \boldsymbol{\mu}_k)^{\top} \Sigma_k^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k).$$

Show that for each i, k , the posterior distribution of $u_{i,k}$ (given $z_i = k$ and $\mathbf{y}_i$) is

$$u_{i,k} \mid (\mathbf{y}_i, z_i = k) \sim \text{Gamma}\left(\frac{\nu_k^{(t)} + p}{2}, \frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2}\right),$$

Hint: multiply the Gaussian likelihood (in u) by the Gamma prior and identify the kernel of a Gamma distribution.

3. (3 pts) Show that hence its posterior moments are

$$w_{i,k}^{(t)} := \mathbb{E}[u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}] = \frac{\nu_k^{(t)} + p}{\nu_k^{(t)} + d_{i,k}^{(t)}},$$

$$\ell_{i,k}^{(t)} := \mathbb{E}[\log u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}] = \psi\left(\frac{\nu_k^{(t)} + p}{2}\right) - \log\left(\frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2}\right).$$

EM Algorithm — M-step — latent-scale representation and closed-form updates.

1. (12 pts) Using the results in E step , prove the closed-form M-step updates.

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \delta_{i,k}^{(t)},$$

$$\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} \mathbf{y}_i}{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)}}$$

and

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t+1)})^\top}{\sum_{i=1}^N \delta_{i,k}^{(t)}}$$

by differentiating the Q -function w.r.t. π_k , $\boldsymbol{\mu}_k$ and Σ_k^{-1} (respectively) and setting the derivatives to zero.

Hints: you can simplify $Q(\theta \mid \theta^{(t)})$ and use $w_{i,k}^{(t)}, \ell_{i,k}^{(t)}, \delta_{i,k}^{(t)}, \theta^{(t)}$ to express it.

2. (3 pts) **degrees of freedom ν_k .**

Starting from the Q -function (after replacing $u_{i,k}$ by its posterior moments $w_{i,k}^{(t)}$ and $\ell_{i,k}^{(t)}$), derive the stationarity equation for ν_k . That is, show that the maximizing $\nu_k^{(t+1)}$ (solution of $\partial Q / \partial \nu_k = 0$) satisfies the scalar equation

$$\log\left(\frac{\nu_k}{2}\right) - \psi\left(\frac{\nu_k}{2}\right) + 1 + \frac{1}{\sum_{i=1}^N \delta_{i,k}^{(t)}} \sum_{i=1}^N \delta_{i,k}^{(t)} (\ell_{i,k}^{(t)} - w_{i,k}^{(t)}) = 0,$$

3. (3 pts) Provide the Newton's method (one-step update) used to solve this equation numerically, i.e.

$$\nu \leftarrow \nu - \frac{f(\nu)}{f'(\nu)},$$

and give explicit expressions for $f(\nu)$ and $f'(\nu)$ (in terms of digamma/trigamma functions).

Hint: set $a = \nu_k/2$ to simplify derivatives of $\log \Gamma(a)$. Use $\frac{d}{da} \log \Gamma(a) = \psi(a)$ and $\frac{d}{da} \psi(a) = \psi'(a)$

Remark

- **Degrees-of-freedom control tail strength:** the parameter ν_k moderates the sensitivity. For fixed $d_{i,k}$, larger ν_k increases the numerator $\nu_k + p$ but also the denominator $\nu_k + d_{i,k}$ — in the limit $\nu_k \rightarrow \infty$ we have $w_{i,k} \rightarrow 1$ and the updates reduce to the Gaussian-mixture EM updates. For small ν_k the down-weighting is stronger: heavy tails mean more robustness.
- **Intuition:** the latent precision $u_{i,k}$ allows each observation to “explain” a different amount of variability. If an observation is unlikely under component k (large Mahalanobis distance), the posterior for $u_{i,k}$ concentrates near small values, effectively increasing the conditional variance for that observation under component k and reducing its influence in parameter estimation. This built-in reweighting is exactly why mixtures of t -components are robust to outliers relative to Gaussian mixtures.

Solution.

E-step: responsibilities and latent-scale posteriors

1. $Q(\theta \mid \theta^{(t)})$ and $H(\theta \mid \theta^{(t)})$

Introduce latent labels $z_i \in \{1, \dots, K\}$ with

$$\Pr(z_i = k) = \pi_k, \quad p(\mathbf{y}_i \mid z_i = k; \theta) = t_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \Sigma_k, \nu_k).$$

The joint density of $(\mathbf{y}_i, z_i)$ under parameter θ is

$$p_\theta(\mathbf{y}_i, z_i = k) = \pi_k t_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \Sigma_k, \nu_k).$$

Thus the complete-data log-likelihood is

$$\begin{aligned}\log p_\theta(\mathbf{Y}, \mathbf{Z}) &= \sum_{i=1}^N \log p_\theta(\mathbf{y}_i, z_i) \\ &= \sum_{i=1}^N \sum_{k=1}^K \mathbf{1}\{z_i = k\} \left[\log \pi_k + \log t_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \Sigma_k, \nu_k) \right],\end{aligned}$$

where $\mathbf{1}\{z_i = k\}$ is the indicator of $z_i = k$. The E-step uses the posterior distribution of Z given the observed data and the current parameters $\theta^{(t)}$:

$$\Pr(z_i = k \mid \mathbf{y}_i; \theta^{(t)}) = \frac{\pi_k^{(t)} t_p(\mathbf{y}_i; \boldsymbol{\mu}_k^{(t)}, \Sigma_k^{(t)}, \nu_k^{(t)})}{\sum_{\ell=1}^K \pi_\ell^{(t)} t_p(\mathbf{y}_i; \boldsymbol{\mu}_\ell^{(t)}, \Sigma_\ell^{(t)}, \nu_\ell^{(t)})} =: \delta_{i,k}^{(t)}.$$

The Q -function. By definition (as in class),

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\mathbf{Z} \mid \mathbf{Y}, \theta^{(t)}} [\log p_\theta(\mathbf{Y}, \mathbf{Z})].$$

Taking expectation of the complete-data log-likelihood with respect to $Z \mid Y, \theta^{(t)}$ simply replaces $\mathbf{1}\{z_i = k\}$ by its posterior mean $\delta_{i,k}^{(t)}$, so

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \sum_{k=1}^K \delta_{i,k}^{(t)} \left[\log \pi_k + \log t_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \Sigma_k, \nu_k) \right]. \text{(3pts)}$$

The H -term. As defined in class,

$$H(\theta \mid \theta^{(t)}) = -\mathbb{E}_{\mathbf{Z} \mid \mathbf{Y}, \theta^{(t)}} [\log p_\theta(\mathbf{Z} \mid \mathbf{Y})],$$

so here the $\log$ involves $p_\theta(\mathbf{Z} \mid \mathbf{Y})$ (the new parameter θ), while the expectation is taken under the old parameter $\theta^{(t)}$. For a given θ , the posterior over Z can be written as

$$p_\theta(\mathbf{Z} \mid \mathbf{Y}) = \prod_{i=1}^N \prod_{k=1}^K (\delta_{i,k}(\theta))^{\mathbf{1}\{z_i=k\}},$$

where

$$\delta_{i,k}(\theta) := \Pr_\theta(z_i = k \mid \mathbf{y}_i) = \frac{\pi_k t_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \Sigma_k, \nu_k)}{\sum_{\ell=1}^K \pi_\ell t_p(\mathbf{y}_i; \boldsymbol{\mu}_\ell, \Sigma_\ell, \nu_\ell)}.$$

Thus

$$\log p_\theta(\mathbf{Z} \mid \mathbf{Y}) = \sum_{i=1}^N \sum_{k=1}^K \mathbf{1}\{z_i = k\} \log \delta_{i,k}(\theta).$$

Taking expectation with respect to $Z \mid Y, \theta^{(t)}$ again replaces $\mathbf{1}\{z_i = k\}$ by $\delta_{i,k}^{(t)}$, giving

$$H(\theta \mid \theta^{(t)}) = -\sum_{i=1}^N \sum_{k=1}^K \delta_{i,k}^{(t)} \log \delta_{i,k}(\theta). \text{(2pts)}$$

2. Posterior of the latent scales $u_{i,k}$

We use the scale-mixture representation (for component k):

$$\mathbf{y}_i \mid (z_i = k, u_{i,k}) \sim \mathcal{N}\left(\boldsymbol{\mu}_k, \frac{\Sigma_k}{u_{i,k}}\right), \quad u_{i,k} \mid (z_i = k) \sim \text{Gamma}\left(\frac{\nu_k}{2}, \frac{\nu_k}{2}\right).$$

Fix an index pair (i, k) and condition on $z_i = k$. We want the posterior

$$p(u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}).$$

By Bayes,

$$p(u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}) \propto p(\mathbf{y}_i \mid z_i = k, u_{i,k}; \theta^{(t)}) p(u_{i,k} \mid z_i = k; \theta^{(t)})(1pt),$$

where we keep only terms depending on $u_{i,k}$.

Likelihood term. Under the multivariate normal,

$$\mathbf{y}_i \mid (z_i = k, u_{i,k}) \sim \mathcal{N}_p\left(\boldsymbol{\mu}_k^{(t)}, \frac{\Sigma_k^{(t)}}{u_{i,k}}\right).$$

Its pdf is

$$p(\mathbf{y}_i \mid z_i = k, u_{i,k}) = \frac{1}{(2\pi)^{p/2} \left| \frac{1}{u_{i,k}} \Sigma_k^{(t)} \right|^{1/2}} \exp\left\{ -\frac{1}{2} (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t)})^\top \left(\frac{1}{u_{i,k}} \Sigma_k^{(t)} \right)^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t)}) \right\}.$$

We simplify the covariance terms:

$$\left| \frac{1}{u_{i,k}} \Sigma_k^{(t)} \right| = \frac{1}{u_{i,k}^p} |\Sigma_k^{(t)}|, \quad \left(\frac{1}{u_{i,k}} \Sigma_k^{(t)} \right)^{-1} = u_{i,k} (\Sigma_k^{(t)})^{-1}.$$

Therefore

$$p(\mathbf{y}_i \mid z_i = k, u_{i,k}) = \frac{u_{i,k}^{p/2}}{(2\pi)^{p/2} |\Sigma_k^{(t)}|^{1/2}} \exp\left\{ -\frac{1}{2} u_{i,k} d_{i,k}^{(t)} \right\} (1pt),$$

where we used the Mahalanobis distance

$$d_{i,k}^{(t)} = d_{i,k}(\boldsymbol{\mu}_k^{(t)}, \Sigma_k^{(t)}) = (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t)})^\top (\Sigma_k^{(t)})^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t)}).$$

For the posterior in $u_{i,k}$, the prefactor $(2\pi)^{-p/2} |\Sigma_k^{(t)}|^{-1/2}$ is a constant and can be ignored.

Prior term. The Gamma prior has pdf

$$p(u_{i,k} \mid z_i = k) = \frac{(\nu_k^{(t)}/2)^{\nu_k^{(t)}/2}}{\Gamma(\nu_k^{(t)}/2)} u_{i,k}^{\nu_k^{(t)}/2-1} \exp\left\{ -\frac{\nu_k^{(t)}}{2} u_{i,k} \right\} (1pt)$$

Again, for the posterior kernel in $u_{i,k}$, only the powers of $u_{i,k}$ and the exponential term matter.

Posterior kernel and identification as Gamma. Multiplying the two kernels we obtain

$$\begin{aligned} p(u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}) &\propto u_{i,k}^{p/2} \exp\left\{ -\frac{1}{2} u_{i,k} d_{i,k}^{(t)} \right\} u_{i,k}^{\nu_k^{(t)}/2-1} \exp\left\{ -\frac{\nu_k^{(t)}}{2} u_{i,k} \right\} \\ &= u_{i,k}^{\frac{\nu_k^{(t)}}{2}-1+\frac{p}{2}} \exp\left\{ -\frac{1}{2} u_{i,k} (\nu_k^{(t)} + d_{i,k}^{(t)}) \right\} \\ &= u_{i,k}^{\frac{\nu_k^{(t)}+p}{2}-1} \exp\left\{ -\frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2} u_{i,k} \right\}. (1pt) \end{aligned}$$

Comparing with the $\text{Gamma}(\alpha, \beta)$ kernel

$$u^{\alpha-1} e^{-\beta u},$$

we identify

$$\alpha = \frac{\nu_k^{(t)} + p}{2}, \quad \beta = \frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2}.$$

Hence

$$u_{i,k} \mid (\mathbf{y}_i, z_i = k; \theta^{(t)}) \sim \text{Gamma}\left(\frac{\nu_k^{(t)} + p}{2}, \frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2}\right),$$

which is the desired result.

3. Posterior moments of $u_{i,k}$

For a $\text{Gamma}(\alpha, \beta)$ (shape–rate) random variable U we have

$$\mathbb{E}[U] = \frac{\alpha}{\beta}(1pt), \quad \mathbb{E}[\log U] = \psi(\alpha) - \log \beta.(2pts)$$

Applying this with

$$\alpha = \frac{\nu_k^{(t)} + p}{2}, \quad \beta = \frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2},$$

gives

$$w_{i,k}^{(t)} := \mathbb{E}[u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}] = \frac{\nu_k^{(t)} + p}{\nu_k^{(t)} + d_{i,k}^{(t)}},$$

$$\ell_{i,k}^{(t)} := \mathbb{E}[\log u_{i,k} \mid \mathbf{y}_i, z_i = k; \theta^{(t)}] = \psi\left(\frac{\nu_k^{(t)} + p}{2}\right) - \log\left(\frac{\nu_k^{(t)} + d_{i,k}^{(t)}}{2}\right),$$

as claimed.

M–step: closed–form updates

We now maximise $Q(\theta \mid \theta^{(t)})$ w.r.t. θ , using the latent–scale representation and replacing $u_{i,k}$ by its posterior expectations.

Augmented Q using latent scales

Given the hierarchical model

$$p(\mathbf{y}_i, z_i, u_{i,1:K}) = \prod_{k=1}^K \left[\pi_k \mathcal{N}_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \frac{\Sigma_k}{u_{i,k}}) \text{Gamma}(u_{i,k}; \frac{\nu_k}{2}, \frac{\nu_k}{2}) \right] \mathbf{1}_{\{z_i=k\}},$$

the complete–data log–likelihood (with u included) is

$$\log p_\theta(\mathbf{Y}, \mathbf{Z}, \mathbf{U}) = \sum_{i=1}^N \sum_{k=1}^K \mathbf{1}_{\{z_i = k\}} \left[\log \pi_k + \log \mathcal{N}_p(\mathbf{y}_i; \boldsymbol{\mu}_k, \frac{\Sigma_k}{u_{i,k}}) + \log \text{Gamma}(u_{i,k}; \frac{\nu_k}{2}, \frac{\nu_k}{2}) \right].$$

The E–step now takes expectation w.r.t. $(Z, U) \mid Y, \theta^{(t)}$. Using $\mathbb{E}[\mathbf{1}_{\{z_i = k\}}] = \delta_{i,k}^{(t)}$ and $\mathbb{E}[u_{i,k} \mid \mathbf{y}_i, z_i = k] = w_{i,k}^{(t)}$, $\mathbb{E}[\log u_{i,k} \mid \mathbf{y}_i, z_i = k] = \ell_{i,k}^{(t)}$, we can write $Q(\theta \mid \theta^{(t)})$ as

$$\begin{aligned} Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \sum_{k=1}^K \delta_{i,k}^{(t)} \left\{ \log \pi_k - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k) \right. \\ \left. + \frac{\nu_k}{2} \log \frac{\nu_k}{2} - \log \Gamma\left(\frac{\nu_k}{2}\right) + \left(\frac{\nu_k}{2} - 1\right) \ell_{i,k}^{(t)} - \frac{\nu_k}{2} w_{i,k}^{(t)} \right\} + \text{const}, \end{aligned}$$

where we have separated terms depending on π_k , $(\boldsymbol{\mu}_k, \Sigma_k)$, and ν_k . For the updates of $\pi_k, \boldsymbol{\mu}_k, \Sigma_k$ we only keep the relevant parts.

1. Update for π_k

The π_k –dependent part of Q is

$$Q_\pi(\theta) = \sum_{k=1}^K \sum_{i=1}^N \delta_{i,k}^{(t)} \log \pi_k,$$

subject to the constraint $\sum_{k=1}^K \pi_k = 1$. Using a Lagrange multiplier λ ,

$$\mathcal{L}(\pi, \lambda) = \sum_{k=1}^K \sum_{i=1}^N \delta_{i,k}^{(t)} \log \pi_k + \lambda \left(1 - \sum_{k=1}^K \pi_k \right).$$

Differentiate w.r.t. π_k :

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^N \delta_{i,k}^{(t)}}{\pi_k} - \lambda.$$

Setting this to zero yields

$$\pi_k = \frac{1}{\lambda} \sum_{i=1}^N \delta_{i,k}^{(t)}.$$

Summing over k and using $\sum_k \pi_k = 1$,

$$1 = \sum_{k=1}^K \pi_k = \frac{1}{\lambda} \sum_{k=1}^K \sum_{i=1}^N \delta_{i,k}^{(t)} = \frac{1}{\lambda} \sum_{i=1}^N \sum_{k=1}^K \delta_{i,k}^{(t)} = \frac{1}{\lambda} \sum_{i=1}^N 1 = \frac{N}{\lambda},$$

so $\lambda = N$ and

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \delta_{i,k}^{(t)}.$$

2. Update for μ_k

The μ_k -dependent part of Q is

$$Q_{\mu, \Sigma}(\theta) \supset -\frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \mu_k)^\top \Sigma_k^{-1} (\mathbf{y}_i - \mu_k).$$

For each k we maximize this w.r.t. μ_k with Σ_k fixed. Using the standard matrix derivative

$$\nabla_{\mu} (\mathbf{y} - \mu)^\top A (\mathbf{y} - \mu) = -2A(\mathbf{y} - \mu),$$

we obtain

$$\begin{aligned} \frac{\partial Q}{\partial \mu_k} &= -\frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (-2) \Sigma_k^{-1} (\mathbf{y}_i - \mu_k) \\ &= \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} \Sigma_k^{-1} (\mathbf{y}_i - \mu_k). \end{aligned}$$

Setting $\partial Q / \partial \mu_k = 0$ and premultiplying by Σ_k gives

$$\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \mu_k) = 0,$$

i.e.

$$\left(\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} \right) \mu_k = \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} \mathbf{y}_i.$$

Therefore

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} \mathbf{y}_i}{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)}}.$$

3. Update for Σ_k

It is convenient to differentiate w.r.t. Σ_k^{-1} . For fixed k , the Σ_k -dependent part of Q is

$$Q_{\Sigma_k}(\theta) = -\frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} \log |\Sigma_k| - \frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k).$$

Using the derivative identities

$$\frac{\partial \log |\Sigma_k|}{\partial \Sigma_k^{-1}} = -\Sigma_k, \quad \frac{\partial}{\partial \Sigma_k^{-1}} \left[(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_k) \right] = (\mathbf{y}_i - \boldsymbol{\mu}_k)(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top,$$

we obtain

$$\begin{aligned} \frac{\partial Q}{\partial \Sigma_k^{-1}} &= -\frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} (-\Sigma_k) - \frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k)(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top \\ &= \frac{1}{2} \left(\sum_{i=1}^N \delta_{i,k}^{(t)} \right) \Sigma_k - \frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k)(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top. \end{aligned}$$

Setting the derivative to zero and multiplying by 2,

$$\left(\sum_{i=1}^N \delta_{i,k}^{(t)} \right) \Sigma_k = \sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k)(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top.$$

Plugging in $\boldsymbol{\mu}_k^{(t+1)}$ for $\boldsymbol{\mu}_k$ gives

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{i,k}^{(t)} w_{i,k}^{(t)} (\mathbf{y}_i - \boldsymbol{\mu}_k^{(t+1)})(\mathbf{y}_i - \boldsymbol{\mu}_k^{(t+1)})^\top}{\sum_{i=1}^N \delta_{i,k}^{(t)}}.$$

4. Stationarity equation for ν_k

Now we focus on the terms of Q that depend on ν_k . From the augmented expression above, collecting all ν_k -dependent terms for a fixed k gives

$$Q_k(\nu_k) = \sum_{i=1}^N \delta_{i,k}^{(t)} \left[\frac{\nu_k}{2} \log \frac{\nu_k}{2} - \log \Gamma\left(\frac{\nu_k}{2}\right) + \left(\frac{\nu_k}{2} - 1\right) \ell_{i,k}^{(t)} - \frac{\nu_k}{2} w_{i,k}^{(t)} \right] + \text{const},$$

where “const” does not depend on ν_k . Let $N_k^{(t)} := \sum_{i=1}^N \delta_{i,k}^{(t)}$. We differentiate Q_k w.r.t. ν_k . It is convenient to set $a = \nu_k/2$. Then

$$\begin{aligned} \frac{d}{d\nu_k} \left(a \log a \right) &= \left(\log a + 1 \right) \frac{da}{d\nu_k} = \frac{1}{2} (\log a + 1), \\ \frac{d}{d\nu_k} \left(-\log \Gamma(a) \right) &= -\psi(a) \frac{da}{d\nu_k} = -\frac{1}{2} \psi(a), \\ \frac{d}{d\nu_k} \left(a \ell_{i,k}^{(t)} \right) &= \frac{1}{2} \ell_{i,k}^{(t)}, \quad \frac{d}{d\nu_k} \left(-a w_{i,k}^{(t)} \right) = -\frac{1}{2} w_{i,k}^{(t)}. \end{aligned}$$

Thus

$$\begin{aligned} \frac{\partial Q_k}{\partial \nu_k} &= \sum_{i=1}^N \delta_{i,k}^{(t)} \left[\frac{1}{2} \left(\log \frac{\nu_k}{2} + 1 \right) - \frac{1}{2} \psi\left(\frac{\nu_k}{2}\right) + \frac{1}{2} \ell_{i,k}^{(t)} - \frac{1}{2} w_{i,k}^{(t)} \right] \\ &= \frac{1}{2} \sum_{i=1}^N \delta_{i,k}^{(t)} \left[\log \frac{\nu_k}{2} - \psi\left(\frac{\nu_k}{2}\right) + 1 + \ell_{i,k}^{(t)} - w_{i,k}^{(t)} \right]. \end{aligned}$$

Setting $\partial Q_k / \partial \nu_k = 0$ and dividing by $N_k^{(t)} / 2 > 0$ yields

$$\log\left(\frac{\nu_k}{2}\right) - \psi\left(\frac{\nu_k}{2}\right) + 1 + \frac{1}{N_k^{(t)}} \sum_{i=1}^N \delta_{i,k}^{(t)} (\ell_{i,k}^{(t)} - w_{i,k}^{(t)}) = 0.$$

Recalling that $N_k^{(t)} = \sum_{i=1}^N \delta_{i,k}^{(t)}$, we obtain exactly the scalar equation in the problem statement:

$$\boxed{\log\left(\frac{\nu_k}{2}\right) - \psi\left(\frac{\nu_k}{2}\right) + 1 + \frac{1}{\sum_{i=1}^N \delta_{i,k}^{(t)}} \sum_{i=1}^N \delta_{i,k}^{(t)} (\ell_{i,k}^{(t)} - w_{i,k}^{(t)}) = 0.}$$

5. Newton update for ν_k

Define

$$f(\nu) := \log\left(\frac{\nu}{2}\right) - \psi\left(\frac{\nu}{2}\right) + 1 + \frac{1}{N_k^{(t)}} \sum_{i=1}^N \delta_{i,k}^{(t)} (\ell_{i,k}^{(t)} - w_{i,k}^{(t)}).$$

The last term is constant with respect to ν during the M-step. To apply Newton's method we need $f'(\nu)$. Using again $a = \nu/2$ and the identities

$$\frac{d}{da} \log \Gamma(a) = \psi(a), \quad \frac{d}{da} \psi(a) = \psi'(a),$$

we compute

$$\begin{aligned} \frac{d}{d\nu} \log\left(\frac{\nu}{2}\right) &= \frac{1}{\nu}, \\ \frac{d}{d\nu} \left[-\psi\left(\frac{\nu}{2}\right) \right] &= -\psi'\left(\frac{\nu}{2}\right) \cdot \frac{1}{2} = -\frac{1}{2} \psi'\left(\frac{\nu}{2}\right), \\ \frac{d}{d\nu} (1) &= 0, \\ \frac{d}{d\nu} \left[\frac{1}{N_k^{(t)}} \sum_{i=1}^N \delta_{i,k}^{(t)} (\ell_{i,k}^{(t)} - w_{i,k}^{(t)}) \right] &= 0. \end{aligned}$$

Hence

$$\boxed{f'(\nu) = \frac{1}{\nu} - \frac{1}{2} \psi'\left(\frac{\nu}{2}\right).}$$

A single Newton step is therefore

$$\boxed{\nu^{\text{new}} \leftarrow \nu^{\text{old}} - \frac{f(\nu^{\text{old}})}{f'(\nu^{\text{old}})}}$$

with $f(\cdot)$ and $f'(\cdot)$ as above, where $\psi(\cdot)$ is the digamma and $\psi'(\cdot)$ the trigamma function. This completes the detailed derivation of the EM algorithm for mixtures of multivariate t -distributions.

Problem 5 (Expectation Maximization Interpretation behind Semi-Supervised Learning) (20 pts)

Given N samples $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^m$ as well as their labels $y_1, \dots, y_N \in \{0, 1, \dots, K\}$. Consider the generative model where each sample $\mathbf{x}_i$ is generated independently according to Gaussian mixture model that depends on the label y_i , as represented by random variable

$$X_i \sim \begin{cases} \sum_{j=1}^K \pi_j \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) & , \text{ if } y_i = 0 \\ \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) & , \text{ if } y_i = k \neq 0 \end{cases}$$

where $\pi_1 + \dots + \pi_K = 1$, and $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, with probability density function

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^m |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

We would like to apply Expectation Maximization algorithm to find the maximum likelihood estimation of parameters $\theta = \{(\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\}_{k=1}^K$.

- (15 pts) Please write down the E-step and M-step and show that the parameters are updated from $\theta^{(t)} = \left\{(\pi_k^{(t)}, \boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)})\right\}_{k=1}^K$ to $\theta^{(t+1)} = \left\{(\pi_k^{(t+1)}, \boldsymbol{\mu}_k^{(t+1)}, \boldsymbol{\Sigma}_k^{(t+1)})\right\}_{k=1}^K$ in the following form:

$$\pi_k^{(t+1)} = \frac{\sum_{i:y_i=0} \delta_{ik}^{(t)}}{\sum_{i:y_i=0} 1}$$

$$\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_{i:y_i=k} \mathbf{x}_i + \sum_{i:y_i=0} \delta_{ik}^{(t)} \mathbf{x}_i}{N_k + \sum_{i:y_i=0} \delta_{ik}^{(t)}}$$

$$\boldsymbol{\Sigma}_k^{(t+1)} = \frac{\sum_{i:y_i=k} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^\top + \sum_{i:y_i=0} \delta_{ik}^{(t)} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^\top}{N_k + \sum_{i:y_i=0} \delta_{ik}^{(t)}}$$

where $N_k = \sum_{i:y_i=k} 1$ is the number of samples in class k . Please show your derivations.

- (5 pts) What is the closed form expression of $\delta_{ik}^{(t)}$? Please show your derivations.

Solution: Given data set $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the likelihood and log-likelihood functions are given by

$$p(S; \theta) = \left(\prod_{i: y_i=0} \sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) \right) \cdot \prod_{k=1}^K \prod_{i: y_i=k} \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

$$\log p(\mathcal{X}; \theta) = \sum_{i: y_i=0} \log \left(\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) \right) + \sum_{k=1}^K \sum_{i: y_i=k} \log \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Denote latent variable $z_i \in \{1, \dots, K\}$ indicating which Gaussian distribution x_i is drawn from.

- Expectation Step (E-step): Compute

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \mathbb{E}_{z_i \mid \mathbf{x}_i, y_i; \theta^{(t)}} [\log p(\mathbf{x}_i, y_i, z_i; \theta)]$$

Posterior prob. dist. of latent variables z_i based on current parameters $\theta^{(t)}$: If $y_i = 0$, then

$$\delta_{ik}^{(t)} = \mathbb{P}[z_i = k \mid \mathbf{x}_i; y_i = 0; \theta^{(t)}] = \frac{p(\mathbf{x}_i, y_i = 0, z_i = k; \theta^{(t)})}{\sum_{j=1}^K p(\mathbf{x}_i, y_i = 0, z_i = j; \theta^{(t)})} = \frac{\pi_k^{(t)} \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)})}{\sum_{j=1}^K \pi_j^{(t)} \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j^{(t)}, \boldsymbol{\Sigma}_j^{(t)})}$$

If $y_i = l \neq 0$, then

$$\delta_{ik}^{(t)} = \mathbb{P}[z_i = k \mid \mathbf{x}_i; y_i = l; \theta^{(t)}] = \begin{cases} 1 & , \text{ if } l = k \\ 0 & , \text{ if } l \neq k \end{cases}$$

- Log-likelihood of parameter θ given data (x_i, y_i) and latent variable z_i :

If $y_i = 0$, then

$$\log p(\mathbf{x}_i, y_i = 0, z_i = k; \theta) = \log \left(\frac{\pi_k}{\sqrt{(2\pi)^m |\boldsymbol{\Sigma}_k|}} \right) - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)$$

If $y_i = l \neq 0$, then

$$\log(\mathbf{x}_i, y_i = k, z_i = l; \theta) = \begin{cases} \log \left(\frac{1}{\sqrt{(2\pi)^m |\boldsymbol{\Sigma}_k|}} \right) - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) & l = k \\ \log(0) & l \neq k \end{cases}$$

Hence

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \sum_{k=1}^K \delta_{ik}^{(t)} \left\{ \log \left(\frac{\pi_k^{I(y_i=0)}}{\sqrt{(2\pi)^m |\boldsymbol{\Sigma}_k|}} \right) - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) \right\}$$

- Maximization Step (M-step): Choose

$$\theta^{(t+1)} = \arg \max_{\theta \in \Theta} Q(\theta \mid \theta^{(t)})$$

Note that $Q(\theta \mid \theta^{(t)})$ takes exactly the same form as in the unsupervised GMM scenario, hence the optimal

solution $\theta^{(t+1)}$ is specified by $\pi_k^{(t+1)} = \frac{\sum_{i: y_i=0} \delta_{ik}^{(t)}}{\sum_{i: y_i=0} 1}$, $\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_{i: y_i=k} \mathbf{x}_i + \sum_{i: y_i=0} \delta_{ik}^{(t)} \mathbf{x}_i}{N_k + \sum_{i: y_i=0} \delta_{ik}^{(t)}}$

$$\boldsymbol{\Sigma}_k^{(t+1)} = \frac{\sum_{i: y_i=k} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^T + \sum_{i: y_i=0} \delta_{ik}^{(t)} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^T}{N_k + \sum_{i: y_i=0} \delta_{ik}^{(t)}}$$